2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N) | 979-8-3503-3086-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICAC3N60023.2023.10541716

Voice Feature Extraction Method Analysis for Speaker Recognition with Degraded Human Voice

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***Abstract*— The human brain possesses millions of neurons that enable easy identification of different speaker voices. However, when a speaker's voice is accompanied by significant background noise, recognizing the speaker becomes challenging. In our experiment, we are focusing on an efficient approach for extracting features that support speaker recognition even with degraded human voice. Mel-Frequency Cepstral Coefficient (MFCC) is a commonly used method for extracting human speech features and is widely utilized in speaker recognition systems. The process of extracting features from MFCC involves dividing the audio into frames and calculating cepstral coefficients for each frame. Each frame in the MFCC exhibits similar spectral power, representing the distinct and shared vocal characteristics of each speaker. The MFCC provides numerical values for these vocal properties, which are effectively employed in identifying the speaker, whether through a Machine Learning (ML) or Artificial Intelligence (AI) enabled system for speaker identification. In this article, we employ a Gaussian Mixture Model (GMM) for speaker classification in scenarios where there is high background noise. Additionally, the experiment investigates the impact of sampling rate on human voice in situations where the signal-to-noise ratio (SNR) is low throughout the speaker recognition process**

Keywords—Mel-Spectrum, voice feature extraction, MFCC, Speaker Recognition, Gaussian Mixture Model

I. INTRODUCTION (*Heading 1*)

Artificial Intelligence (AI) and Machine Learning (ML) not only automate tasks, providing humans with extra time, but they also contribute to the identification of neurological issues through speech analysis. Nowadays, numerous electronic devices are incorporating voice-based instructions and can even detect neurological changes in human speech. Multiple applications are using human voice for authentication purpose. For these applications, speaker classification based on voice is crucial. Speaker recognition is relatively straightforward when there is no background noise. However, accuracy becomes more challenging when dealing with degraded human voice. Typically, users provide voice inputs in noisy environments such as traffic noise or nature-inspired noises like air friction. These practical noises significantly reduce the accuracy of speaker classification. While humans can easily identify individuals based on their voices, machines face greater difficulty. Artificial Intelligence (AI) plays a significant role in achieving reliable speaker recognition, especially when high background noise is present, resulting in a low Signal-to-Noise Ratio (SNR). In the speaker recognition process, Machine Learning (ML) algorithms are commonly used, but prior to the ML stage, feature extraction and pre-processing are crucial stages, as illustrated in Fig-1

Fig. 1. Speaker Recognition Base Process.

Pre-processing involves cleaning the input voice signal by removing known noise. This can be accomplished using Band Pass Filters (BPF) or Low Pass Filters (LPF). These filters effectively eliminate specific frequencies associated with the noise, ensuring a cleaner input signal.

Feature extraction stage helps to find different types of voice features which are helpful in recognition of speaker. These extracted human voice properties are used in learning and pattern matching stages to recognize speaker. Different machine learning algorithms like Gaussian Mixer Model (GMM)[1][2][3] and Hidden Markov Model (HMM)[4] is helpful in identification of speaker. Support Vector Machine (SVM) is commonly worked for learning and pattern matching [3][5][6][7][8].

In speaker recognition, when there is high traffic noise accompanying the human voice, the feature extraction process may not yield distinct features that can effectively classify the speaker. Traffic noise, as well as air friction noise, can occur both inside and outside a house. For instance, traffic noise can penetrate into a house if it is located near a road, and air friction noise can be present indoors or outdoors. When the level of air friction noise exceeds the speaker's voice, the accuracy of the speaker recognition system decreases. The accuracy of the system (speaker identification system) is highest when no background noise is present. However, this is an ideal scenario with no background noise.

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The Signal-to-Noise Ratio (SNR) will always higher when either the background noise is low compared to the voice signal or the power of the speaker voice signal is high as compared to the power of the background noise signal. The SNR represents the ratio of the power of the voice signal (Ps) to the power of the background noise (Pn).

$$SNR = \frac{Ps}{Pn}$$

Human voice signals are not constant but rather exhibit variability. The characteristics of human speech can change due to various factors such as emotions, speaking style, health conditions, and environmental influences. Hence, power of the voice signal in logarithmic decibel scale(dB) can expressed as:

$$Ps(dB) = 10log_{10}\, Ps$$

Similarly, power from noise signal can be expressed in logarithmic decibel scale(dB) as

$$Pn(dB) = 10log_{10}\, Pn$$

Signal to noise ratio (SNR) can also be articulated in dB.

$$SNR(dB) = 10log_{10}(Ps/Pn)$$

OR

$$SNR(dB) = [Ps(dB) - Pn(dB)]$$

Ps consider human voice signal power in absence of any noise and Pn consider as noise power. When we have human voice signal then that signal may have background noise. So to calculate Ps, it is required to remove noise power first from original human signal[9].

$$SNR(dB) = 10log_{10}(\frac{Ps-Pn}{Pn})$$

The Signal-to-Noise Ratio compare the power of the desired signal (human voice signal) to the power of the background noise. When the power of the human voice signal is higher than the power of the background noise, the SNR will be positive. This indicates a favourable SNR where the voice signal is relatively stronger compared to the noise.

Conversely, when the power of the human voice signal is lower than the power of the background noise, the SNR will be negative. This indicates a less favourable signal-to-noise ratio where the noise is relatively stronger compared to the voice signal. Negative SNR values are often represented in decibels (dB) to quantify the extent of the signal degradation caused by the background noise[9].

II. Related Work

The Gaussian Mixture Model is generally utilized as an efficient model in speaker recognition tasks. In a previous experiment [10], it was demonstrated that utilizing a reduced Mel-Frequency Cepstral Coefficient (MFCC) feature vector along with power spectrum density can enhance the overall speaker verification process. By incorporating these additional features, the accuracy of speaker recognition can be improved. Furthermore, speaker gender classification can be accomplished based on the features extracted from the speaker's voice [11]. Male voices typically exhibit a frequency range of 100-146Hz, which is consistently lower than the frequency range of female voices, typically found between 188-221Hz. By analyzing these frequency characteristics, it becomes possible to differentiate between male and female speakers.

In their work, S. Park, Y. Park, A. Nasridinov and J. Lee [12] employed a similar approach to differentiate between genders and allocate person IDs for real-time minutes of meetings, utilizing speaker information. They utilized the "Text-To-Speech" API algorithm provided by Google to convert the spoken voice into text and generate a runtime meeting report. This approach allows for automated transcription and analysis of meeting discussions, facilitating the creation of detailed summaries or records.

Convolutional Neural Networks (CNNs) have gained significant popularity in the field of speaker recognition [13]. CNNs are particularly effective when combined with Siamese networks [14], which are neural networks designed to compare and measure similarity between inputs. The application of CNNs in speaker recognition is especially advantageous when dealing with degraded signals that have undergone Mel-Frequency Cepstral Coefficient (MFCC) and Linear Predictive Coding (LPC) processing [15]. This suggests that CNNs are a preferred choice for speaker recognition tasks, particularly in scenarios where the input signals have undergone specific feature extraction techniques.

Both Support Vector Machines (SVM) and Gaussian Mixture Models (GMM) are widely recognized as efficient models for speaker recognition. Each model has its own set of advantages and disadvantages. In [3], the authors delve into the specific advantages of GMM and SVM in detail. Furthermore, they propose an approach to combine the strengths of both models, aiming to leverage the advantages of each approach and enhance the speaker identification system performance. By integrating the strengths of GMM and SVM, it is possible to achieve improved accuracy and robustness in speaker recognition tasks. The paper likely provides further insights into the specific advantages of each model and how they can be effectively combined.

Indeed, when the human voice is accompanied by limited noise, it becomes feasible for machines to identify the speaker from their voice. In the work by N. V. Tahliramani and N. Bhatt [16], they described that machines generally recognize human speech features using Mel-Frequency Cepstral Coefficients (MFCC) during the training phase. MFCC is a commonly used technique for extracting relevant features from human speech. By training a machine learning model with MFCC features, it becomes possible for the machine to learn and recognize the distinct characteristics of individual speakers. This enables the machine to identify speakers accurately, given a voice sample with minimal background noise. The research likely delves into the details of how MFCC features are utilized in training machines for speaker recognition purposes.

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There are multiple other methods also which is used for voice feature extraction like Residual Phase Cepstrum Coefficients (RPCC)[17][18] and Glottal feature [17][19].

III. Feature Extraction Method Comparision

There are multiple way to classify speaker. GMM, HMM and SVM are the three popular classification technique for speaker classification [23][24][25].

GMM is a statistical model used for representing complex probability distributions. The model assumes that the data is generated from a mixture of Gaussian (normal) distributions. Each Gaussian distribution in the mixture represents a component, and the combination of these components forms the overall probability distribution. GMMs are commonly used for modeling the acoustic features extracted from speech signals. Each speaker is represented by a GMM, where the components capture the variations in the speech characteristics of that speaker. During the training phase, the GMM parameters are estimated using a large set of speech samples from each speaker. The GMM can then be used to compute the likelihood of a given speech sample belonging to a particular speaker. GMMs offer flexibility and robustness in modeling complex distributions and have been widely utilized in various applications, including speech and speaker recognition, as well as other pattern recognition tasks. GMM generates different types of clusters, and each cluster is modelled as per different Gaussian distribution[26]

HMM is also a statistical model that is widely used for modelling consecutive data, where the primary process is presumed to be a Markov processd with hidden states. HMMs are particularly suitable for tasks that involve temporal or sequential dependencies, such as speech recognition, natural language processing, and handwriting recognition. In an HMM, the underlying process consists of a set of states that are not directly observable (hidden states). Instead, the model emits observable symbols or outputs based on the current hidden state. The transitions between hidden states are governed by transition probabilities, while the emission of symbols from each state is determined by emission probabilities. HMMs are characterized by three main components.

- State Transition Probabilities: This define the probability of transitioning from one hidden state to another at each time step. They are typically represented by a transition matrix.
- Emission Probabilities: These probabilities determine the likelihood of emitting specific observable symbols or outputs from each hidden state. They are often represented by an emission matrix or set of emission probabilities.
- Initial State Distribution: This distribution specifies the probabilities of starting in each hidden state at the beginning of the sequence.

SVM is a generally used for classification of speaker and comes under in category of supervised machine learning algorithm. SVMs are particularly effective in solving binary classification problems, but they can also be extended to handle multi-class classification. The main idea behind SVM is to identify an ideal hyperplane that splits data points belonging to distinct classes with the heaviest possible margin. The hyperplane is determined by support vectors, which are a subset of training data points that lie closest to the decision boundary.

In this article we have used GMM because SVM and HMM is not much effective when human voice sample duration is too less. For feature extraction we have used Mel-Frequency Cepstral Coefficient (MFCC). MFCC value varies when we change sampling rate. In our experiment we have seen sampling rate effects in time domain voice signal. Sampling rate also impact power spectrum and feature matrix of MFCC.

IV. Experiment and Results

Industries typically utilize speaker recognition applications for biometric purposes, specifically when training human voice samples recorded in quieter environments. These environments are characterized by minimal background noise, resulting in a high signal-to-noise ratio (SNR). However, when testing samples gathered from noisier surroundings, such as factories, traffic, or other environmental factors, the SNR significantly decreases due to the substantial background noise present.

In our experiment we have used most silent sample as training sample. As per table I, voice sample ID 1 is used as training sample as it has high SNR as compared to testing samples. We used low SNR sample as testing sample in range of 2dB to 10dB

TABLE I. Voice Samples

Voice Sample ID	Sample SNR Range	
	Sample Use	***SNR(dB)***
1	Training	11 to 12
2	Testing	9 to 10
3	Testing	8 to 9
4	Testing	3 to 5
5	Testing	2 to 3

In our experiment we have used MFCC for feature extraction and GMM for speaker classification through learning and pattern matching. We have performed two different test, Test-1 and Test-2. In Test-1 we have used sampling rate(SR) 22050 and in Test-2 we have used SR as 44100. In both the test hardware and other MFCC parameters are remain same. We have used same dataset in both the tests. In our scenario, we did not mix human voice with separate noise sources. Instead, we conducted the recordings in the same room where both the human voice and background noise were present simultaneously. We employed an identical instrument to generate varying intensities of air friction noise, resulting in different signal-to-noise ratio (SNR) levels for the samples. It is important to note that throughout the testing phase, we consistently used the same voice recorder for all voice samples. Table-II is showing both the test results.

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TABLE II. EXPERIMENT RESULTS WITH SAMPLING RATE 22050/44100

Voice Sample ID	Test-1(22050) and Test-2(SR-44100) Results			
	Sample Use	*SNR(dB)*	*Test-1 Speaker Classified?*	*Test-2 Speaker Classified?*
1	Training	11 to 12	Training Sample	Training Sample
2	Testing	9 to 10	**No**	**Yes**
3	Testing	8 to 9	**No**	**Yes**
4	Testing	3 to 5	No	No
5	Testing	2 to 3	No	No

Table-II shows experiment results and we observed that sample ID 2 and 3 are classified correctly with high sampling rate.

V. CONCLUSION

In section IV of our discussion, we explored the influence of sampling rate on the process of Mel-frequency cepstral coefficient (MFCC) extraction and its impact on speaker recognition. Through our observations, we discovered that performing the feature extraction at a lower sampling rate leads to difficulties in correctly classifying speakers within a low SNR range of 8 to 10dB. However, as we increase the sampling rate, the number of samples per frame significantly increases, resulting in sharper and more distinct feature characteristics. This enhancement effectively supports the overall speaker recognition process, particularly when employing Gaussian mixture models (GMM).

REFERENCES

[1] O. Büyük and L. M. Arslan, "Age identification from voice using feed-forward deep neural networks," 2018 26th Signal Processing and Communications Applications Conference (SIU), Izmir, 2018, pp. 1-4

[2] Z. Weng, L. Li and D. Guo, "Speaker recognition using weighted dynamic MFCC based on GMM," 2010 International Conference on Anti-Counterfeiting, Security and Identification, 2010, pp. 285-288

[3] H. C. Bao and Z. C. Juan, "The research of speaker recognition based on GMM and SVM," 2012 International Conference on System Science and Engineering (ICSSE), 2012, pp. 373-375

[4] Y. Wei, "Adaptive Speaker Recognition Based on Hidden Markov Model Parameter Optimization," in IEEE Access, vol. 8, pp. 34942-34948

[5] S. Cumani and P. Laface, "Large-Scale Training of Pairwise Support Vector Machines for Speaker Recognition," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 22, no. 11, pp. 1590-1600, Nov. 2014

[6] R. Mardhotillah, B. Dirgantoro and C. Setianingsih, "Speaker Recognition for Digital Forensic Audio Analysis using Support Vector Machine," 2020 3rd International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), 2020

[7] R. M. Lexușan, "Comparative study regarding characteristic features of the human voice," 2015 7th International Conference on Electronics, Computers and Artificial Intelligence (ECAI), Bucharest, 2015, pp. WSD-1-WSD-4

[8] B. K. Baniya, J. Lee and Z. Li (2014), " Audio feature reduction and analysis for automatic music genre classification", In IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 457-462

[9] P. Papadopoulos, A. Tsiartas, J. Gibson and S. Narayanan, "A supervised signal-to-noise ratio estimation of speech signals," 2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2014, pp. 8237-8241

[10] R. Chakroun, L. Beltaïfa Zouari, M. Frikha and A. Ben Hamida, "Improving text-independent speaker recognition with GMM," 2016 2nd International Conference on Advanced Technologies for Signal and Image Processing (ATSIP), 2016, pp. 693-696

[11] W. Meiniar, F. A. Afrida, A. Irmasari, A. Mukti and D. Astharini, "Human voice filtering with band-stop filter design in MATLAB," 2017 International Conference on Broadband Communication, Wireless Sensors and Powering (BCWSP), Jakarta, 2017, pp. 1-4

[12] S. Park, Y. Park, A. Nasridinov and J. Lee, "A Person Identification Method in CUG Using Voice Pitch Analysis," 2014 IEEE Fourth International Conference on Big Data and Cloud Computing, Sydney, NSW, 2014, pp. 765-766

[13] A. H. Meftah, H. Mathkour, S. Kerrache and Y. A. Alotaibi, "Speaker Identification in Different Emotional States in Arabic and English," in IEEE Access, vol. 8, pp. 60070-60083, 2020

[14] N. Gupta and S. Jain, "Speaker Identification Based Proxy Attendance Detection System," 2019 International Conference on Signal Processing and Communication (ICSC), NOIDA, India, 2019, pp. 175-179

[15] A. Chowdhury and A. Ross, "Fusing MFCC and LPC Features Using 1D Triplet CNN for Speaker Recognition in Severely Degraded Audio Signals," in IEEE Transactions on Information Forensics and Security, vol. 15, pp. 1616-1629, 2020

[16] N. V. Tahliramani and N. Bhatt, "Performance Analysis of Speaker Identification System With and Without Spoofing Attack of Voice Conversion," 2018 2nd International Conference on Micro-Electronics and Telecommunication Engineering (ICMETE), Ghaziabad, India, 2018, pp. 130-135

[17] T. Thomas, S. V, N. V. Sobhana and S. G. Koolagudi, "Speaker Recognition in Emotional Environment using Excitation Features," 2020 Third International Conference on Advances in Electronics, Computers and Communications (ICAECC), 2020, pp. 1-6

[18] J. Wang and M. T. Johnson, "Physiologically-motivated feature extraction for speaker identification," 2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2014, pp. 1690-1694

[19] S. Ostrogonac, M. Sečujski, D. Knezevic and S. Suzić, "Extraction of glottal features for speaker recognition," 2013 IEEE 9th International Conference on Computational Cybernetics (ICCC), 2013, pp. 369-373

[20] F. Eyben et al., "The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing," in IEEE Transactions on Affective Computing, vol. 7, no. 2, pp. 190-202, 1 April-June 2016

[21] M. Sadeghi and H. Marvi, "Optimal MFCC features extraction by differential evolution algorithm for speaker recognition," 2017 3rd Iranian Conference on Intelligent Systems and Signal Processing (ICSPIS), 2017, pp. 169-173

[22] A. Winursito, R. Hidayat and A. Bejo, "Improvement of MFCC feature extraction accuracy using PCA in Indonesian speech recognition," 2018 International Conference on Information and Communications Technology (ICOIACT), 2018, pp. 379-383

[23] P. Pujol, S. Pol, C. Nadeu, A. Hagen and H. Bourlard, "Comparison and combination of features in a hybrid HMM/MLP and a HMM/GMM speech recognition system," in IEEE Transactions on Speech and Audio Processing, vol. 13, no. 1, pp. 14-22, Jan. 2005,

[24] H. Zeinali, H. Sameti and L. Burget, "HMM-Based Phrase-Independent i-Vector Extractor for Text-Dependent Speaker Verification," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 25, no. 7, pp. 1421-1435, July 2017

[25] W. M. Campbell, "A SVM/HMM system for speaker recognition," 2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings. (ICASSP '03)., 2003,

[26] A. K. Devika, M. G. Sumithra and A. K. Deepika, "A fuzzy-GMM classifier for multilingual speaker identification," 2014 International Conference on Communication and Signal Processing, 2014.

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